

Activity <2.2>

<CONTROL STRUCTURES (part 1)>

Course Code: CPE 007

Program: Computer Engineering

Course Title: Programming logic and design

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6. Output

Example pseudocode:

While there are more items on my shopping list

Purchase next item and cross it off my list

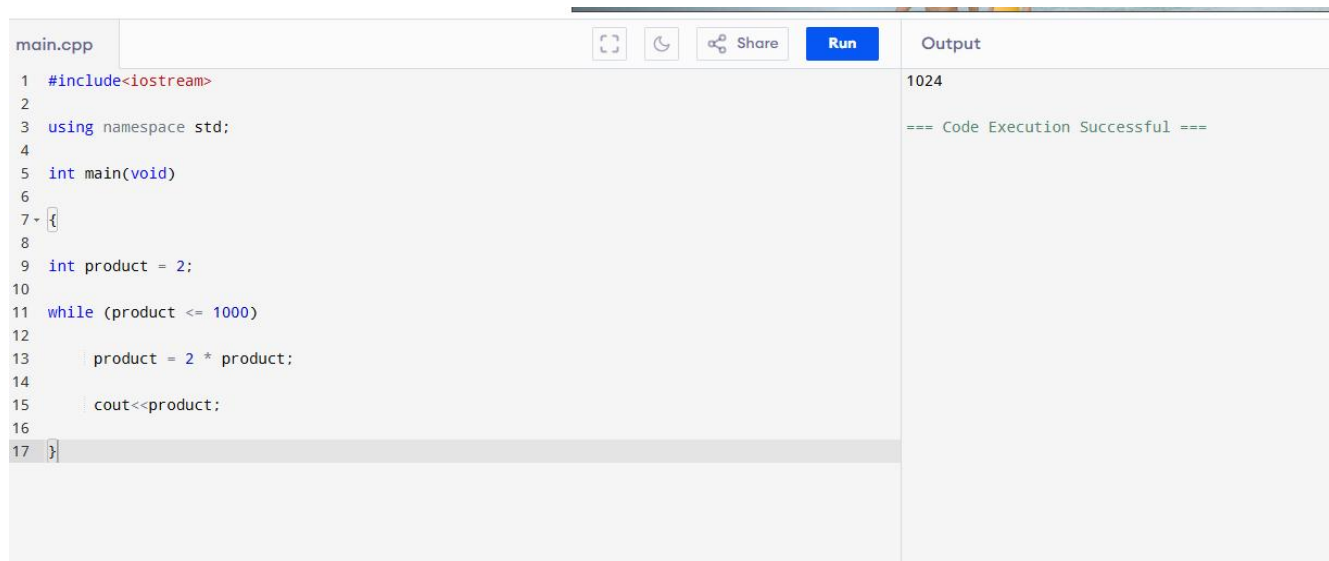
Example program:

```
#include<iostream>

using namespace std;

int main(void)
{
    int product = 2;
    while (product <= 1000)
        product = 2 * product;
    cout<<product;
}
```

What do you think is the output of this program? = 1024



The screenshot shows a C++ IDE with a file named 'main.cpp'. The code is as follows:

```
1 #include<iostream>
2
3 using namespace std;
4
5 int main(void)
6 {
7     int product = 2;
8     while (product <= 1000)
9         product = 2 * product;
10    cout<<product;
11 }
```

The IDE has a 'Run' button and a 'Share' button. The output window on the right shows the result of the program execution:

```
1024
=== Code Execution Successful ===
```

Exercise 1

Using the following pseudocode the program can be as follows:

Set total to zero

Set grade counter to one

While grade counter is less than or equal to ten

Input the next grade

Add the grade into the total

Add one to the grade counter

Set the class average to the total divided by ten

Print the class average

main.cpp	Run	Output
<pre>1 #include <iostream> 2 using namespace std; 3 4 int main() { 5 int total = 0; 6 int gradeCounter = 1; 7 int grade; 8 double average; 9 10 11 while (gradeCounter <= 10) { 12 cout << "Enter grade " << gradeCounter << ": "; 13 cin >> grade; 14 15 total += grade; 16 gradeCounter++; 17 } 18 19 20 average = static_cast<double>(total) / 10; 21 22 cout << "Class average: " << average << endl; 23 24 return 0; 25 }</pre>	  	<pre>Enter grade 1: 76 Enter grade 2: 65 Enter grade 3: 45 Enter grade 4: 66 Enter grade 5: 75 Enter grade 6: 88 Enter grade 7: 67 Enter grade 8: 54 Enter grade 9: 89 Enter grade 10: 87 Class average: 71.2 === Code Execution Successful ===</pre>

7. Supplementary Activity

1. Using conditional statements (if-else statements), write a program that asks a user for a number and prints out if it is an even or an odd number.

odd

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 4 int main() { 5 int num; 6 7 cout << "Enter a number: "; 8 cin >> num; 9 10 if (num % 2 == 0) { 11 cout << num << " is an even number." << endl; 12 } else { 13 cout << num << " is an odd number." << endl; 14 } 15 16 return 0; 17 }</pre>	<pre>Enter a number: 3 3 is an odd number. === Code Execution Successful ===</pre>

even

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 4 int main() { 5 int num; 6 7 cout << "Enter a number: "; 8 cin >> num; 9 10 if (num % 2 == 0) { 11 cout << num << " is an even number." << endl; 12 } else { 13 cout << num << " is an odd number." << endl; 14 } 15 16 return 0; 17 } 18 19 20 21</pre>	<pre>Enter a number: 2 2 is an even number. === Code Execution Successful ===</pre>

2. Using conditional statements, write a program that computes for 10 percent fare discount of a senior citizen and 8 percent fare discount of a student. There will be no discount if not a senior citizen and not a student. The user will be asked to enter age. The minimum fare is 9 pesos.

Senior discount:

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 4 int main() { 5 int age; 6 const double fare = 9.0; 7 double discount = 0.0, totalFare; 8 9 cout << "Enter your age: "; 10 cin >> age; 11 12 if (age >= 60) { 13 discount = fare * 0.10; 14 cout << "Senior citizen discount applied." << endl; 15 } 16 else if (age >= 18 && age <= 22) { 17 discount = fare * 0.08; 18 cout << "Student discount applied." << endl; 19 } 20 else { 21 cout << "No discount applied." << endl; 22 } 23 24 totalFare = fare - discount; 25 cout << "Total Fare: ₱" << totalFare << endl; 26 27 return 0; 28 }</pre>	<pre>Enter your age: 60 Senior citizen discount applied. Total Fare: ₱8.1 === Code Execution Successful ===</pre>

No discount:

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 4 int main() { 5 int age; 6 const double fare = 9.0; 7 double discount = 0.0, totalFare; 8 9 cout << "Enter your age: "; 10 cin >> age; 11 12 if (age >= 60) { 13 discount = fare * 0.10; 14 cout << "Senior citizen discount applied." << endl; 15 } 16 else if (age >= 18 && age <= 22) { 17 discount = fare * 0.08; 18 cout << "Student discount applied." << endl; 19 } 20 else { 21 cout << "No discount applied." << endl; 22 } 23 24 totalFare = fare - discount; 25 cout << "Total Fare: ₱" << totalFare << endl; 26 27 return 0; 28 }</pre>	<pre>Enter your age: 9 No discount applied. Total Fare: ₱9 === Code Execution Successful ===</pre>

3. Case Study: Sentinel Controlled Repetition. Given the following pseudocode, create a program that will implement a sentinel controlled repetition. For example, you can use (-1) as the sentinel value. You can use Problem 1 as your reference.

main.cpp	Run Share <pre>1 #include <iostream> 2 using namespace std; 3 4 int main() { 5 int grade; 6 int total = 0; 7 int counter = 0; 8 double average; 9 10 cout << "Enter a grade (-1 to end): "; 11 cin >> grade; 12 13 14 while (grade != -1) { 15 total += grade; 16 counter++; 17 18 cout << "Enter next grade (-1 to end): "; 19 cin >> grade; 20 } 21 22 23 if (counter != 0) { 24 average = static_cast<double>(total) / counter; 25 cout << "Average grade: " << average << endl; 26 } else { 27 cout << "No grades were entered." << endl; 28 } 29 30 return 0; 31 }</pre>	Output
		<pre>Enter a grade (-1 to end): 98 Enter next grade (-1 to end): 7 Enter next grade (-1 to end): 78 Enter next grade (-1 to end): 76 Enter next grade (-1 to end): 88 Enter next grade (-1 to end): 97 Enter next grade (-1 to end): 87 Enter next grade (-1 to end): 99 Enter next grade (-1 to end): -1 Average grade: 78.75 === Code Execution Successful ===</pre>

No grades were entered:

main.cpp	Run Share <pre>1 #include <iostream> 2 using namespace std; 3 4 int main() { 5 int grade; 6 int total = 0; 7 int counter = 0; 8 double average; 9 10 cout << "Enter a grade (-1 to end): "; 11 cin >> grade; 12 13 14 while (grade != -1) { 15 total += grade; 16 counter++; 17 18 cout << "Enter next grade (-1 to end): "; 19 cin >> grade; 20 } 21 22 -1 23 if (counter != 0) { 24 average = static_cast<double>(total) / counter; 25 cout << "Average grade: " << average << endl; 26 } else { 27 cout << "No grades were entered." << endl; 28 } 29 30 return 0; 31 }</pre>	Output
		<pre>Enter a grade (-1 to end): -1 No grades were entered. === Code Execution Successful ===</pre>

Pseudocode:

Initialize total to zero

Initialize counter to zero

Input the first grade

While the user has not as yet entered the sentinel

Add this grade into the running total

Add one to the grade counter

Input the next grade (possibly the sentinel)

If the counter is not equal to zero

Set the average to the total divided by the counter

Print the average

else

Print "No grades were entered"

8. Conclusion

I learned a lot because of this activity, such as how to use if/else and while, and how to use them properly. And now I also understand how and where the sentinel is used.

9. Assessment Rubric